

Department Overview Department Name: Department of Electronic Engineering Faculty: Faculty of Information & Communication Technology (FICT) Accreditation: The program is fully accredited by the Pakistan Engineering Council (PEC). • Core Technological Focus: Robotics, Artificial Intelligence, Digital Manufacturing, Automated and Electric Vehicles, Information Security, Wireless Networks, and Nano-Technologies. History & Impact: Established in 2003, the department focuses on meeting the shifting technological needs of industry and society, playing a primary role in technical education within Balochistan. Career Fields: Graduates are employed across diverse sectors, including national and international research institutions, telecom companies, banking, automotive, textile, biomedical, petrochemical, manufacturing, freelancing, and entrepreneurship.

Program Educational Objectives (PEOs) The department aims to produce electronic engineers who can:

1. PEO-1: Contribute their knowledge for the betterment of humanity in general and society in particular.
2. PEO-2: Impart their analytical and innovation skills for the enhancement of research, development, and production of Electronic Systems.
3. PEO-3: Work efficiently in teams or individually, integrating effectively in diverse cultures and environments.

Bachelor of Science (BS) Program

Admission Requirements F.Sc Pre-Engineering: Completed from any recognized board or equivalent with a minimum of 60% marks. Diploma of Associate Engineer (DAE): Completed in a relevant field with at least 60% marks.

Degree Requirements Total Credit Hours: 136 Total Courses: 45 Minimum CGPA: ≥ 2.0

Curriculum Plan (Program Schema by Semester) Note: Credit hour listings follow the format (Theory + Lab).

1st Semester • HUM-163 / MATHP-118 | Functional English / Calculus and Analytical Geometry | Prerequisite: NIL/NIL | Credit Hours: 3+0 / 3+0 • EE-112 & EE-112L | Linear Circuit Analysis | Prerequisite: NIL | Credit Hours: 3+1 • PHY-205 & PHY-205L | Applied Physics | Prerequisite: NIL | Credit Hours: 2+1 • CS-123 / ELE-102L | Applications of ICT / Workshop Practice | Prerequisite: NIL/NIL | Credit Hours: 2+1 / 0+1

2nd Semester • HUM-268 / HUM-101 | Communication Skills / Islamic Studies | Prerequisite: NIL/NIL | Credit Hours: 3+0 / 2+0 • EE-208 & EE-208L | Electronic Devices & Circuits | Prerequisite: NIL | Credit Hours: 3+1 • ENGG-217L / HUM-102 | Engineering Drawing / Pakistan Studies | Prerequisite: NIL/NIL | Credit Hours: 0+1 / 2+0 • MATHP-111 | Linear Algebra | Prerequisite: NIL | Credit Hours: 3+0

3rd Semester • STAT-403 / MATHA-214 | Probability & Random Variables / Differential Equations | Prerequisite: NIL / Calculus, Geometry, & Linear Alg | Credit Hours: 3+0 / 3+0 • EE-110 & EE-110L | Digital Logic Design | Prerequisite: NIL | Credit Hours: 3+1 • EE-204 & EE-204L | Instrumentation & Measurements | Prerequisite: Electronic Devices & Circuits | Credit Hours: 2+1 • EE-109 & EE-109L | Electronic Circuit Design | Prerequisite: NIL | Credit Hours: 3+1

4th Semester • HUM-276 / MATHA-213 | TRW & PS / Complex Variables and Transform | Prerequisite: NIL/NIL | Credit Hours: 3+0 / 3+0 • EE-491 & EE-491L | Integrated Electronics Analog | Prerequisite: Electronic Circuit Design | Credit Hours: 2+1 • EE-201 & EE-201L |

Electrical Network Analysis | Prerequisite: Linear Circuit Analysis | Credit Hours: 3+1 • CE-324 & CE-324L | Microprocessors and Microcontrollers | Prerequisite: Digital Logic Design | Credit Hours: 3+1

5th Semester • EE-324 & EE-324L | Electrical Machines | Prerequisite: NIL | Credit Hours: 3+1 • STAT-403 | Signal & Systems | Prerequisite: NIL | Credit Hours: 2+1 • EE-241 & EE-241L | Control Systems | Prerequisite: Linear Algebra | Credit Hours: 3+1 • CS-212 & CS-212L | Object Oriented Programming for Engineers | Prerequisite: NIL | Credit Hours: 3+1 • TE-301 & TE-301L | Data Communication & Networks | Prerequisite: Communication Systems | Credit Hours: 2+1

6th Semester • EE-421 | Power Electronics | Prerequisite: Electrical Machines | Credit Hours: 3+1 • MATHA-216 & L / EE-XX | Numerical Analysis & Lab / IDEE-I | Prerequisite: NIL | Credit Hours: 2+1 / 3+0 • TE-314 & TE-314L | Communication Systems | Prerequisite: Signals and Systems | Credit Hours: 3+1 • MGMT-XX | Management Elective-I | Prerequisite: NIL | Credit Hours: 3+0

7th Semester • EE-XX / EE-XX | Elective-I / Elective-II | Prerequisite: NIL/NIL | Credit Hours: 3+0 / 3+0 • MGMT-XX | Management Elective-II | Prerequisite: NIL | Credit Hours: 3+0 • EE-441 & EE441L | Industrial Electronics | Prerequisite: NIL | Credit Hours: 3+1 • XX-XX / EE-481L | Social Science Elective-I / Final Year Project Part-I | Prerequisite: NIL/NIL | Credit Hours: 2+0 / 0+3

8th Semester • EE-XX / EE-XX | Elective-III / Elective-IV | Prerequisite: NIL/NIL | Credit Hours: 3+0 / 3+0 • TE-412 & TE-412L | Digital Signal Processing | Prerequisite: Signals & Systems | Credit Hours: 2+1 • XX-XX / XX-XX | Social Science Elective-II / IDEE-II | Prerequisite: NIL/NIL | Credit Hours: 2+0 / 3+0 • EE-482L | Final Year Project Part-II | Prerequisite: Final Year Project Part-I | Credit Hours: 0+3

Master of Science (MS) Program

Admission Requirements Academic Background: BE/BS degree in Electronic, Electrical, Computer, or Telecommunication Engineering from an HEC-recognized institution. Academic Performance: Minimum of 60% marks under the annual grading system or a minimum CGPA of 2.5 out of 4.0. Standardized Testing: A valid GAT General score with at least a 50% cumulative result.

Degree Requirements Total Credit Hours: 30 Total Courses: 8 Minimum Academic Standing: CGPA $\geq$ 2.5